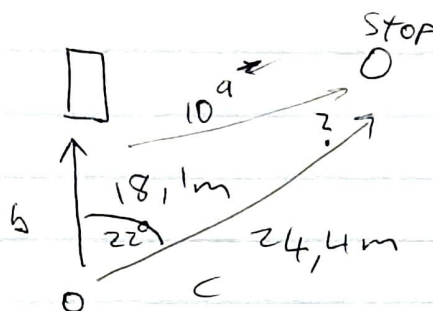


Q1



$$a^2 = b^2 + c^2 - 2ab \cos A$$

$$\begin{aligned} 1.1) a^2 &= (18,1)^2 + (24,4)^2 - 2(18,1)(24,4) \cos(22) \\ &= 100,397 \\ &= 10,0 \text{ m} \end{aligned}$$

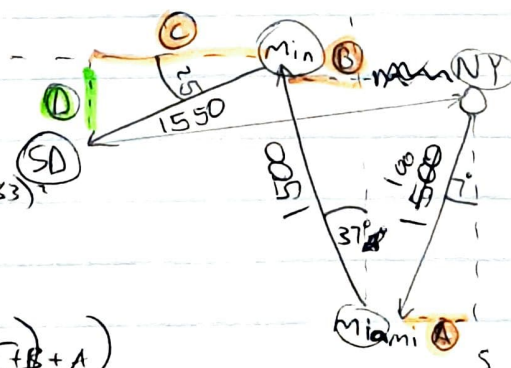
$$\begin{aligned} b^2 &= c^2 + a^2 - 2ac \cos \hat{B} \\ (18,1)^2 &= (24,4)^2 + (10)^2 - 2(10)(24,4) \cos \hat{B} \\ 367,75 &= 20(24,4) \cos \hat{B} \\ 0,7535 &= \cos \hat{B} \\ \therefore \hat{B} &= 41,1^\circ \end{aligned}$$

1.2) a)

$$\begin{aligned} D_{\text{net}} &= \sqrt{(A+B+C)^2 + D^2} \\ &= \sqrt{(2441,556)^2 + (655,0583)^2} \\ &= 2527,9 \text{ mi} \end{aligned}$$

$$\begin{aligned} \theta &= \tan^{-1} \left(\frac{C}{D} \right) = \tan^{-1} \left(\frac{655,0583}{2441,556} \right) \\ &= \tan^{-1} \left(\frac{1404,777}{655,0583} \right) \\ &= 25,15^\circ \rightarrow 15,0^\circ \end{aligned}$$

$\Rightarrow 2527,9 \text{ mi } 15,0^\circ \text{ South of West}$



$$\begin{aligned} A &= \sin(7) = \frac{A}{1100} = 134,056 \text{ mi} \\ B &= \sin(37) = \frac{B}{1500} = 902,723 \text{ mi} \\ C &= \cos(25) = \frac{C}{1550} = 1404,777 \text{ mi} \\ D &= \sin(25) = \frac{D}{1550} = 655,0583 \end{aligned}$$

$$b) \text{ speed} = \frac{D}{t}$$

$$= \frac{1100 + 1500 + 1550}{43200}$$

$$= 96,1 \times 10^{-3} \text{ m/h}$$

$$t = 17280 + 8640 + 17280$$

$$= 43200 \text{ h}$$

$$1.3) \rho = \frac{m}{V}$$

$$1 \text{ pound} = 0,453592 \text{ kg}$$

$$1 \text{ inch} = 25,4 \text{ mm}$$

$$12 \text{ inch} = 1 \text{ Foot}$$

$$\therefore 1 \text{ foot} = 0,3048 \text{ m}$$

$$156 \times \frac{0,453592}{(0,3048 \times 12)^3} = 2498,9 \text{ kg/m}^3$$

$$156 \times \frac{0,453592 \text{ kg}}{(0,3048 \text{ m})^3} = 2498,9 \text{ kg/m}^3$$

(Q2)

Elevator

$$x = v_{ix} \Delta t$$

$$\Delta y = v_{iy} \Delta t$$

Bolt

$$y_f = y_i + v_{iy} \Delta t + \frac{1}{2} g \Delta t^2$$

$$\Delta y = v_{iy} \Delta t + \frac{1}{2} g \Delta t^2$$

$$v_{fy} \Delta t = v_{iy} \Delta t + \frac{1}{2} g \Delta t^2$$

$$\frac{v_{fy} \Delta t - v_{iy} \Delta t}{\frac{1}{2} g} = \Delta t$$

$$0 = -v_{fy} \Delta t + v_{iy} \Delta t + \frac{1}{2} g \Delta t^2$$

$$0 = \Delta t (-v_{fy} + v_{iy} + \frac{1}{2} g \Delta t)$$

$$t = 0 \quad \text{or}$$

$$\frac{v_{fy} - v_{iy}}{\frac{1}{2} g} = t$$

b) Bolt

$$v_f = v_i + at$$

$$0 = 5,4 + (-9,8) t$$

$$t = 0,55 \text{ s}$$

$$\text{floors} = \frac{1,49}{3}$$

$$= 0,50 \text{ floors.}$$

$$c) \Delta y = \frac{1}{2} (v_i + v_f) t$$

$$= \frac{1}{2} (-5,4)(0,55)$$

$$\therefore \Delta y = -1,487$$

$$= 1,49$$

Q3

a) $\vec{a} = a_x \hat{i} + a_y \hat{j}$

$$\vec{v}(t) = (a_x t) \hat{i} + a_y \hat{j}$$

$$\vec{x}(t) = \left(\frac{a_x}{2}\right)t^2 \hat{i} + (a_y t) \hat{j}$$

b) i) $\vec{x}(t) = 4,25 \times 10^{14} t^2 \hat{i} + 1,43 \times 10^{15} t \hat{j}$

ii) $\vec{v}(t) = (8,5 \times 10^{14})t \hat{i} + (1,43 \times 10^{15}) \hat{j}$

iii) ~~speed = $\frac{d}{dt}$~~ speed = $|\vec{v}|$
 $= 7,28 \times 10^{15} \text{ m/s}$

iv) $\theta = \tan^{-1} \left(\frac{1,43 \times 10^{15}}{8,5 \times 10^{14}} \right)$

$$= 59,27^\circ$$